BAIO Learning System

I PUC BIOLOGY

MID-TERM PREPARATION

Score High in 2026

- Important Topics
- Diagrams to Practise
- Exam Tips



Narendra Sir
Biology Lecturer
Founder/CEO, BAIO Learning System

How to use this video

- Pause on every topic and **write the answer** in your notebook
- Then check it with the slide and correct mistakes
- Draw every diagram **as given by Narendra Sir**
- Revise the key facts again just before the exam

CHAPTER

1

The Living World

Narendra Sir · Founder/CEO, BAIO

Key facts

- **Species** is the basic unit of classification
- Scientific name: genus with a capital letter, species with a small letter, in italics or underlined — *Mangifera indica* (mango)

Binomial nomenclature

- Naming an organism with **two words**: generic name + specific epithet
- Example: *Homo sapiens*
- Proposed by **Carolus Linnaeus**

CHAPTER

2

Biological Classification

Narendra Sir · Founder/CEO, BAIO

Key facts

- Fungal cell wall is made of **chitin**
- Red rot of sugarcane is caused by a **fungus**, *Colletotrichum falcatum*

Kingdom Monera

- Prokaryotic, unicellular; no membrane-bound organelles
- Cell wall of **peptidoglycan**
- Most abundant; found even in hot springs, deserts, snow, deep oceans
- Autotrophic (photo / chemosynthetic) or heterotrophic
- Reproduce mainly by fission
- Examples: *E. coli*, *Nostoc*, *Anabaena*, *Mycoplasma*

Whittaker's five kingdoms (1969)

Kingdom	Key features	Examples
Monera	Prokaryotic, unicellular, peptidoglycan wall	Bacteria, cyanobacteria
Protista	Unicellular eukaryotes, aquatic	*Chlamydomonas*, *Paramecium*
Fungi	Chitin wall, heterotrophic	*Mucor*, *Agaricus*, yeast
Plantae	Cellulose wall, autotrophic	Algae to flowering plants
Animalia	No cell wall, holozoic	Sponges to mammals

Criteria: cell structure, body organisation, nutrition, reproduction, phylogeny

Five kingdoms — merits and demerits

Merits

- Prokaryotes separated from eukaryotes
- Unicellular eukaryotes grouped in Protista
- Fungi separated from plants
- Reflects phylogeny better than two kingdoms

Demerits

- Protista is a mixed group (*Chlamydomonas* with *Paramecium*)
- Archaebacteria and eubacteria together in Monera
- Viruses, viroids and lichens not placed
- Photosynthetic cyanobacteria kept away from plants

CHAPTER

3

Plant Kingdom

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Key facts

- Bryophytes = **amphibians of the plant kingdom**
- *Cycas* is a **living fossil** among gymnosperms
- *Chlorella*: protein-rich alga used by space travellers

Economic importance of algae

- Food: *Porphyra*, *Laminaria*, *Sargassum*; supplements *Chlorella*, *Spirulina*
- Algin (brown algae) and carrageen (red algae) used commercially
- Agar from *Gelidium* and *Gracilaria* — for culture media, ice-creams, jellies
- Algae fix about half of the total CO_2 on earth

Classes of algae

Class	Common name	Pigments	Stored food	Cell wall
Chlorophyceae	Green algae	Chl a, b	Starch	Cellulose, pectose
Phaeophyceae	Brown algae	Chl a, c, fucoxanthin	Mannitol, laminarin	Cellulose, algin
Rhodophyceae	Red algae	Chl a, d, phycoerythrin	Floridean starch	Cellulose, pectin, polysulphate esters

CHAPTER

4

Animal Kingdom

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Key facts

- **Water vascular system** — Echinodermata
- Mammals are **homoiothermous** (warm-blooded)
- Sponges belong to phylum **Porifera**

Chordates and non-chordates

Chordates	Non-chordates
Notochord present	Notochord absent
CNS dorsal, hollow, single	CNS ventral, solid, double
Pharyngeal gill slits present	Gill slits absent
Heart ventral	Heart dorsal (if present)
Post-anal tail present	Post-anal tail absent

Phylum Arthropoda

- Largest phylum — about two-thirds of all named species
- Bilateral, triploblastic, segmented, coelomate; chitinous exoskeleton
- **Jointed appendages**; body = head, thorax, abdomen
- Respiration: gills, book gills, book lungs, tracheae; **open circulation**
- Excretion by **Malpighian tubules**; antennae, compound eyes, statocysts
- Examples: *Apis*, *Bombyx*, *Laccifer*, *Anopheles*, *Locusta*, *Limulus*

Identify the phylum

Key characters	Phylum
Proboscis gland, collar and trunk	Hemichordata
Comb plates, bioluminescence	Ctenophora
Jointed appendages, Malpighian tubules	Arthropoda
Notochord, hollow nerve cord, gill slits	Chordata
Calcareous shell, head-foot-visceral mass, radula	Mollusca

CHAPTER

5

Morphology of Flowering Plants

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Key facts and venation

- Edible part of mango: fleshy **mesocarp**
- Venation = arrangement of veins in the lamina
- **Reticulate** — network of veinlets (dicots)
- **Parallel** — veins run parallel (monocots)

Phyllotaxy

- Arrangement of leaves on the stem
- **Alternate** — one leaf per node: china rose, mustard, sunflower
- **Opposite** — a pair per node: *Calotropis*, guava
- **Whorled** — more than two per node: *Alstonia*

Placentation

- Arrangement of ovules within the ovary
- **Marginal** — pea
- **Axile** — china rose, tomato, lemon
- **Parietal** — mustard, *Argemone* (false septum)
- **Free central** — *Dianthus*, *Primrose*
- **Basal** — sunflower, marigold

CHAPTER

6

Anatomy of Flowering Plants

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Key facts

- Root hairs are outgrowths of **epidermal cells**
- Stomata are regulated by bean-shaped **guard cells**

Open and closed vascular bundles

Open	Closed
Cambium present	Cambium absent
Secondary growth possible	No secondary growth
Dicot stems	Monocot stems
Bundle sheath usually absent	Sclerenchymatous bundle sheath

CHAPTER

8

Cell: The Unit of Life

Narendra Sir · Founder/CEO, BAIO

Key facts

- **Mitochondria** — powerhouses of the cell (ATP)
- Plant cell wall is made of **cellulose**
- Ribosomes are made in the **nucleolus**
- **Mesosomes**: infoldings of bacterial plasma membrane — help in wall formation, DNA replication, respiration, secretion

Plant cell and animal cell

Plant cell	Animal cell
Cellulose cell wall	No cell wall
Plastids present	No plastids
One large central vacuole	Vacuoles small or absent
Centrioles usually absent	Centrioles present

Diagram — Plant cell

✏ Draw the diagram as given by Narendra Sir

Labels to include: Cell wall · plasma membrane · cytoplasm · nucleus · nucleolus · chloroplast · mitochondrion · central vacuole · tonoplast · Golgi apparatus · ER · ribosomes · plasmodesmata

Fluid mosaic model

- Proposed by **Singer and Nicolson (1972)**
- **Lipid bilayer**: polar heads outside, non-polar tails inside
- **Quasi-fluid** — lipids and proteins move sideways
- Proteins: **peripheral** (surface) and **integral** (buried)
- RBC membrane: about 52% protein, 40% lipid
- Selectively permeable — passive and active transport

Diagram — Fluid mosaic model

✏ Draw the diagram as given by Narendra Sir

Labels to include: Lipid bilayer · polar head · non-polar tails · integral protein · peripheral protein · cholesterol · glycoprotein · glycolipid

Mitochondrion

- Double membrane-bound; inner membrane folds into **cristae**
- Intermembrane space and **matrix**
- **F_0–F_1 particles** on cristae make ATP
- Matrix: circular DNA, RNA, 70S ribosomes, Krebs cycle enzymes
- Site of aerobic respiration; semi-autonomous

Diagram — Mitochondrion

✏ Draw the diagram as given by Narendra Sir

Labels to include: Outer membrane · inner membrane · cristae · intermembrane space · matrix · F_o–F_1 particles · circular DNA · 70S ribosomes

Structures and their organelles

Structure	Found in
Cisternae	Golgi apparatus
Cristae	Mitochondrion
Thylakoids	Chloroplast
Nucleolus	Nucleus
Tonoplast	Vacuole

CHAPTER

9

Biomolecules

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Key facts

- **Purines:** adenine, guanine · **Pyrimidines:** cytosine, thymine, uracil
- Proteins are polymers of **amino acids**
- **Polysaccharides**: chains of monosaccharides joined by glycosidic bonds — cellulose, starch, glycogen, inulin, chitin

Structure of a nucleotide

- **Nitrogenous base** + **pentose sugar** + **phosphate**
- Base + sugar (N-glycosidic bond at 1′C) = **nucleoside**
- Nucleoside + phosphate (phosphoester bond at 5′C) = **nucleotide**
- Nucleotides join by 3′–5′ **phosphodiester bonds**

Factors affecting enzyme activity

- **Temperature** — optimum; low = inactive (reversible), high = denatured
- **pH** — each enzyme has an optimum pH
- **Substrate concentration** — velocity rises, then reaches V_{max}; K_m = [S] at ½ V_{max}
- **Activators and inhibitors** — malonate competitively inhibits succinic dehydrogenase

CHAPTER

10

Cell Cycle and Cell Division

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Key facts

- **Metaphase**: chromosomes at the equator, spindle fibres on kinetochores
- **Crossing over**: exchange between non-sister chromatids, by recombinase
- Crossing over occurs in **pachytene** of Prophase I

Mitosis and meiosis

Mitosis	Meiosis
Somatic cells	Reproductive cells
One division, 2 cells	Two divisions, 4 cells
2n → 2n	2n → n
No crossing over	Crossing over in Prophase I
Identical daughter cells	Genetically different cells

Phases of the cell cycle

- **G_1** — cell grows; no DNA replication
- **S** — DNA replicates (2C → 4C)
- **G_2** — proteins made for mitosis
- **M phase** — karyokinesis (P, M, A, T) + cytokinesis
- **G_0** — quiescent stage; cells exit from G_1

Diagram — Cell cycle

✏ Draw the diagram as given by Narendra Sir

Labels to include: G_1 · S · G_2 · M phase · G_0 · interphase · direction arrows

Diagrams to practise

1. Plant cell
2. Animal cell
3. Fluid mosaic model
4. Mitochondrion
5. Chloroplast
6. Nucleus
7. Bacterial cell
8. Cell cycle
9. Stages of mitosis
10. Types of phyllotaxy
11. Venation
12. Types of placentation
13. Stomata with guard cells
14. Open & closed vascular bundles
15. Structure of a nucleotide

Narendra Sir's tips — how to study

- Read a topic, close the book, write it from memory
- Practise every diagram at least three times
- Learn examples and scientific names with each topic
- Revise differences as tables
- Use the **BAIO Learning System** app daily for MCQs and flashcards

Narendra Sir's tips — in the exam hall

- Read the whole paper first; plan your choices
- Part A: only the **first answer** is evaluated
- Match the marks: 2 marks = 2 sentences, 5 marks = 200–250 words
- Write in points, underline key words, give examples
- **Unlabelled diagrams get no marks**
- Keep the last 10 minutes to check



All the best!

Work hard, stay confident, score high.

Narendra Sir

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Practise more on the BAIO Learning System app